

VIGO AP, Spain

WMO: 080450

Lat: 42.2387N

Lon: 8.6237W

Elev: 855

StdP: 14.25

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 34.5	(c) 36.8	(d) 25.0	(e) 19.7	(f) 42.6	(g) 28.5	(h) 23.1	(i) 42.2	(j) 25.0	(k) 53.5	(l) 22.3	(m) 53.1	(n) 5.4	(o) 230	(p) 0.464

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 17.7	(c) 87.6	(d) 68.8	(e) 84.1	(f) 67.4	(g) 80.7	(h) 66.1	(i) 70.9	(j) 83.9	(k) 69.1	(l) 80.4	(m) 67.5	(n) 77.3	(o) 7.1	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.6	101.0	75.3	65.1	96.0	73.0	64.1	92.4	71.5	35.2	84.2	33.6	80.5	32.3	77.5	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 19.8	(b) 17.0	(c) 15.0	(e) 31.1	(f) 94.1	(g) 2.4	(h) 4.1	(i) 29.4	(j) 97.1	(k) 28.0	(l) 99.5	(m) 26.6	(n) 101.8	(o) 24.8	(p) 104.8		
(a) 19.8	(b) 17.0	(c) 15.0	(e) 29.2	(f) 74.1	(g) 2.6	(h) 2.8	(i) 27.3	(j) 76.1	(k) 25.8	(l) 77.8	(m) 24.4	(n) 79.4	(o) 22.5	(p) 81.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.7	48.1	49.3	53.1	55.0	59.0	64.4	67.4	68.2	65.4	60.3	52.6	49.2		
	DBStd	8.81	4.40	4.67	5.54	6.01	5.85	5.76	5.29	5.01	5.06	5.30	4.85	4.73		
	HDD50	295	88	62	28	15	1	0	0	0	0	2	27	72		
	HDD65	3050	523	438	369	305	206	79	31	19	55	163	373	489		
	CDD50	3116	31	43	125	166	281	433	539	563	461	322	104	48		
	CDD65	395	0	0	1	5	21	62	104	117	67	18	0	0		
	CDH74	2896	0	0	12	46	168	523	818	853	393	83	0	0		
	CDH80	873	0	0	0	4	33	163	272	295	94	12	0	0		
	WSAvg	6.6	6.8	6.7	7.1	7.2	7.0	6.7	6.4	6.4	5.9	6.2	6.4	6.9		
	PrecAvg	59.1	7.3	4.6	4.8	5.2	4.3	1.8	1.3	1.8	3.8	8.3	8.4	7.3		
(15) Precipitation	PrecMax	83.0	14.3	12.8	24.4	15.6	10.6	5.6	3.6	5.4	11.3	14.4	19.1	23.0		
	PrecMin	37.4	0.8	0.0	0.0	1.7	0.7	0.0	0.0	0.2	0.4	0.0	0.4	0.3		
	PrecStd	11.4	4.2	3.5	4.9	3.7	2.6	1.4	1.1	1.3	2.6	3.8	5.0	4.9		
	PrecStd	11.4	4.2	3.5	4.9	3.7	2.6	1.4	1.1	1.3	2.6	3.8	5.0	4.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.8	67.4	75.5	79.1	84.0	89.7	92.2	93.3	87.8	81.4	69.5	63.1		
		MCWB	52.7	55.1	57.9	61.4	64.4	69.3	71.1	71.7	67.9	63.8	59.7	54.8		
	2%	DB	58.4	62.3	70.5	74.1	78.8	84.7	86.5	87.1	82.3	75.6	64.7	59.6		
		MCWB	53.8	52.6	55.6	59.0	62.9	67.0	69.3	68.6	66.0	62.9	58.6	55.6		
	5%	DB	56.4	58.8	66.2	69.5	74.1	80.4	82.5	82.7	78.7	71.4	62.0	57.7		
		MCWB	53.3	51.3	54.2	57.2	61.7	65.7	68.7	67.3	64.8	61.3	57.5	54.7		
	10%	DB	55.0	56.0	61.9	65.1	69.6	75.5	78.4	78.4	74.6	67.9	59.4	56.2		
		MCWB	52.3	51.0	52.6	55.4	59.5	63.7	66.8	66.0	63.8	60.6	55.9	53.5		
	0.4%	WB	57.6	57.1	60.0	62.9	68.1	71.9	73.7	74.5	70.0	67.7	63.1	59.3		
		MCDB	58.6	62.3	71.4	74.8	79.9	85.3	87.7	89.0	83.1	74.8	65.0	60.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	55.8	55.2	57.4	60.3	64.7	68.9	70.9	70.7	67.7	65.4	60.7	57.4		
		MCDB	56.8	58.3	65.8	71.0	75.0	81.2	84.0	83.0	78.4	70.6	63.4	58.4		
	5%	WB	54.3	53.8	55.7	58.3	62.4	66.7	68.9	68.7	66.1	63.7	58.7	55.9		
		MCDB	55.5	56.7	61.8	67.0	72.0	77.6	80.3	78.6	74.8	68.1	60.7	57.1		
	10%	WB	52.9	52.3	54.2	56.4	60.3	64.6	67.1	67.1	64.6	62.2	56.9	54.1		
		MCDB	54.4	55.0	59.5	63.3	67.7	74.0	76.9	75.6	72.1	66.1	58.9	55.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.3	13.3	14.6	14.8	15.3	16.4	17.1	17.7	16.5	13.9	11.8	11.3		
		MCDBR	12.7	17.5	22.1	22.1	22.7	25.1	25.2	25.5	22.9	19.9	14.7	11.9		
		MCWBR	9.1	10.3	11.8	11.8	11.8	12.6	12.0	11.3	10.3	10.2	9.1	8.9		
		MCDBR	9.0	12.9	17.6	19.3	20.2	22.3	22.8	21.5	19.1	14.9	10.8	9.2		
	5% WB	MCWBR	8.0	9.2	10.5	11.0	11.1	11.9	11.4	10.6	9.6	8.8	8.2	8.3		
		MCWBR	8.0	9.2	10.5	11.0	11.1	11.9	11.4	10.6	9.6	8.8	8.2	8.3		
(40) Clear-Sky Solar Irradiance	taub		0.326	0.333	0.366	0.366	0.372	0.374	0.364	0.371	0.361	0.357	0.332	0.321		
	taud		2.504	2.470	2.384	2.392	2.397	2.406	2.432	2.410	2.448	2.482	2.508	2.513		
	Ebn at Noon		260	277	279	286	286	285	286	281	277	264	255	251		
	Edh at Noon		23	28	34	36	37	37	36	35	31	27	23	21		
(44) All-Sky Solar Radiation	RadAvg		515	829	1177	1577	1912	2120	2136	1933	1511	948	599	467		
	RadStd		88	129	192	176	185	145	128	135	103	129	103	74		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+61

Nomenclature: See separate page